

0/24 Questions Answered

hw8-diff-means-regression-teen-pregnancy

Student Name

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Q1

1 Point

Note: There are three options for homework 8, that differ by data topic, and you only need to complete one of the three assignments.

There have been a lot of different factors proposed to be related to teenage pregnancy, such as sex education, family history, etc. High school dropout is more common among teenage mothers, so researchers are interested in identifying factors related to teenage pregnancy to identify potential interventions, such as increasing access to contraception, as well as where those interventions should be targeted.

One researcher was interested in investigating if teen pregnancy rates are related to the percent poverty for US states, because increased poverty may be related to decreased access to contraceptives or funding for sex education. The researcher pulled data from the CDC on `teen pregnancy rates` (number of births per 1,000 females aged 15–19) and on `percent poverty` (percent of people in poverty in the past 12 months) by US state in 2022 reported in the [American Community Survey](#).

We will use this data to investigate differences in states' teen pregnancy rate based on their percent poverty.

	Teen Pregnancy $\bar{x}$	Teen Pregnancy s	n
Above average poverty	16.5300	4.6809	30
Below average poverty	9.9857	3.1052	21

In 2022 the [US official poverty rate](#) was 11.5%, so we will defined a state to have above average poverty if they have a percent poverty above 11.5%, and we will similarly define states to have below average poverty if they have a percent poverty of or below 11.5%. Let

- μ_1 = mean teen pregnancy rate for states with above average poverty rate
- μ_2 = mean teen pregnancy rate for states with below average poverty rate

Q1.1**1 Point**

Is this an example of paired or unpaired data?

Unpaired

Paired

Save Answer

Q2**7 Points**

We will conduct a hypothesis test to investigate if there is evidence of a difference in mean teen pregnancy rate between states above and below the national poverty average, at a significance level of 0.05.

Q2.1**1 Point**

Select the appropriate hypotheses.

$$H_0 : \mu_1 - \mu_2 \neq 0$$

$$H_1 : \mu_1 - \mu_2 = 0$$

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_1 : \mu_1 - \mu_2 \neq 0$$

$$H_0 : \mu_2 - \mu_1 \neq 0$$

$$H_1 : \mu_2 - \mu_1 = 0$$

$$H_0 : \mu_2 - \mu_1 = 0$$

$$H_1 : \mu_2 - \mu_1 \neq 0$$

Save Answer

Q2.2**1 Point**

What is the point estimate for the difference in population mean teen pregnancy rates between states above and below the national poverty rate? Provide your final answer, rounding to four decimal places (make sure to keep the order of groups in mind).

Save Answer

Q2.3**1 Point**

What is the standard error of the point estimate? Round to four decimal places. Upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer****Q2.4****1 Point**

Our next step is to calculate the p-value for our test. We know the difference of means is approximately t-distributed, but what are the degrees of freedom of this t-distribution?

$$df = 30 - 1 = 29$$

$$df = 21 - 1 = 20$$

$$df = 30 + 21 = 51$$

Save Answer**Q2.5****1 Point**

Calculate the p-value. Write your answer with four significant digits (recall you can enter 2.351×10^{-7} as `2.351e-7`).

 Please select file(s)**Select file(s)****Save Answer**

Q2.6**2 Points**

You should have found a very small p-value, much smaller than the significance level. Specify whether we reject the null hypothesis in support of the alternative hypothesis or if we fail to reject the null hypothesis, and then provide a conclusion in context based on the p-value.

[Save Answer](#)

Q3**3 Points**

Now we are not just interested in investigating if there is a difference, we actually want to estimate the difference in means. Let's calculate a 95% confidence interval to express our uncertainty in our point estimate.

Q3.1**1 Point**

What is the critical value for a 95% confidence interval for a mean? Round to four decimal places.

[Save Answer](#)**Q3.2****1 Point**

What is the lower bound of the 95% confidence interval for the difference of means? Round to four decimal places. Upload a picture of your work.

 Please select file(s)[Select file\(s\)](#)[Save Answer](#)

Q3.3**1 Point**

What is the upper bound of the 95% confidence interval for the difference of means? Round to four decimal places. Upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer**

Q4**5 Points**

In this homework we conducted a hypothesis test to test if there was evidence of a difference between the two population means. Then we calculated a confidence interval to estimate the difference between the population means.

What if instead we just want to study the relationship between teen pregnancy rate and percent poverty for US states, where percent poverty is a continuous variable (instead of discretized into above and below)?

Let's review simple linear regression:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Identify each component of the equation by selecting its definition.

Q4.1**1 Point**

x_i is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

Q4.2**1 Point** y_i is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

Q4.3**1 Point** ϵ_i is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

Q4.4**1 Point** β_1 is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

Q4.5**1 Point** β_0 is:

the numeric variable we want to learn about (response variable)

a variable we think will help explain the response variable (explanatory variable)

the y-intercept for the linear relationship between the response and explanatory variables

the slope for the linear relationship between between the response and explanatory variables

the residual/error

Save Answer

Q5**4 Points**

In our case, the variable we want to learn about is teen pregnancy rate and percent poverty is the explanatory variable. We wanted to understand if there is a linear relationship between teen pregnancy rate and percent poverty, at a significance level of 0.05.

Q5.1**1 Point**

What is the correct set of hypotheses to test if mean state teen pregnancy rate linearly varies by percent poverty?

$$H_0 : \beta_0 = 0$$

$$H_a : \beta_0 \neq 0$$

$$H_0 : \beta_0 \neq 0$$

$$H_a : \beta_0 = 0$$

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

$$H_0 : \beta_1 \neq 0$$

$$H_a : \beta_1 = 0$$

Save Answer

Q5.2

1 Point

Below we fit the simple linear regression model and show the output.

```
> pregnancy_model <- lm(pregnancy_rate ~ percent_poverty, data = pregnancy)
> broom::tidy(pregnancy_model)
# A tibble: 2 x 5
  term          estimate std.error statistic    p.value
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)   -6.49      1.98     -3.27 1.96e- 3
2 percent_poverty  1.64      0.157    10.5 4.33e-14
```

What is the p-value for our test? Write your answer with three significant digits (recall you can enter 2.35×10^{-6} as `2.35e-6`).

Q5.3

1 Point

What would be an appropriate conclusion?

We reject the null hypothesis. We found evidence of a linear relationship between teen pregnancy rates and percent poverty for US states.

We fail to reject the null hypothesis. We found evidence of a linear relationship between teen pregnancy rates and percent poverty for US states.

We reject the null hypothesis. We did not find evidence of a linear relationship between teen pregnancy rates and percent poverty for US states.

We failed to reject the null hypothesis. We did not find evidence of a linear relationship between teen pregnancy rates and percent poverty for US states.

Q5.4

1 Point

We found evidence of a linear relationship, but let's provide an estimate with uncertainty of how much we expect teen pregnancy rate to differ for a one unit increase in a state's percent poverty.

```
> confint(pregnancy_model)
              2.5 %      97.5 %
(Intercept) -10.481895 -2.504298
percent_poverty  1.326104  1.956302
```

What is the 95% confidence interval for the slope of the regression? Round to four decimal places.

Lower bound:

Upper bound:

Save Answer

Q6**3 Points**

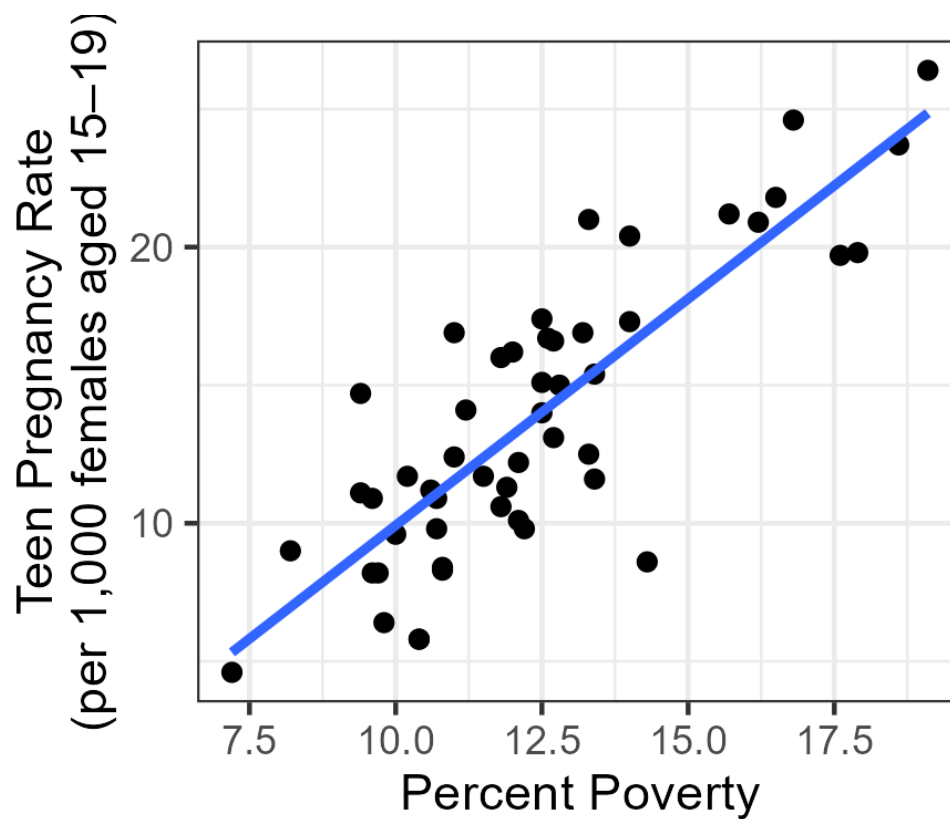
Two key things that linear regression is helpful for are:

- Investigating the expected difference in the response variable when the explanatory variable changes (through the estimate and confidence interval of β_1). *This is what we did in question 5.*
- Predicting the value for the response variable, given a value for the explanatory variable. *This is what we will do in question 6.*

Q6.1

1 Point

The linear regression line we fit is plotted below over the data.



Specify our estimated equation for the regression line (hint: look at the output shown in question 5.2):

[Save Answer](#)

Q6.2**1 Point**

Say a state has a percent poverty of 15. What would we estimate their teen pregnancy rate to be? Enter your numeric answered and upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer****Q6.3****1 Point**

Decreasing a state's percent poverty will decrease its teen pregnancy rate.

True

False

We can't say from this analysis

Save Answer

Q7**1 Point**

The following two questions are related to the homework choice you made in choosing to do this assignment instead of the other two options. There are no correct answers, this question is only being graded for completeness (note that answering "I felt like it" or "it was interesting" is an insufficient response).

Please explain in detail what led you to choose this topic, teen pregnancy, to investigate.

What was it about the other two options (dog lifespan and typing speed that led you not to choose them? Please explain in detail.

Save Answer

Q8**0 Points**

- Have you done the homework by yourself, with a peer(s), or used tools like ChatGPT? If you do not acknowledge any help from a person or ChatGPT, we will consider this cheating.

- If you provided help to a peer(s) and/or got help from your peers(s) please explain briefly what this entailed.

- If you used ChatGPT, include the prompts you used and explain how ChatGPT supported your learning.

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